Ahasan Kabir | Ph.D. Student in Computer Science

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Education

University of Central Florida — **Orlando, USA**
Ph.D. in Computer Science — *Aug 2025 – Present*
Advisor: Dr. Qian Lou

Chittagong University of Engineering and Technology — **Chattogram, Bangladesh**
B.Sc. in Computer Science and Engineering — *Jan 2018 – Jul 2023*

Research Interests

LLM serving, AI infrastructure, and efficient inference systems.

Work Experience

University of Central Florida — **Orlando, FL**
Graduate Research Assistant — *Aug 2025 – Present*

- Working on LLM systems research, including scalable inference, serving infrastructure, and performance optimization across multiple heterogeneous GPUs and workloads.
- Developed **HW-Router**, a hardware-aware routing framework that integrates GPU-level telemetry (queue length, KV-cache usage, utilization, TTFT/TPOT history) with a lightweight latency predictor to make SLO-driven routing decisions. Achieved **3.4–3.9× lower end-to-end latency**, **46–48 pp higher SLO attainment**, and **6–8× lower GPU load skew** compared to CARROT and IRT.

Kinetik Healthcare Solutions, Inc. — **New York, USA**
Software Engineer (Remote) — *May 2023 – Jul 2025*

- Built Kinetik's medical transportation iOS app from scratch in Swift; published on the App Store (link).
- Implemented trip booking and tracking flows, core UI logic, and key app functionality.
- Contributed to backend services using TypeScript, Node.js, NestJS, and gRPC.

Publications & Manuscripts

[1]: **HW-Router: Hardware-Aware Routing for Scalable Multi-LLM Serving** (first author).
Accepted at DAC 2026.

[2]: **InfraMind: Infrastructure-Aware Multi-Agent Orchestration** (first author).
arXiv preprint arXiv:2606.11440, 2026 (link)

[3]: **BEmoLexBERT: A Hybrid Model for Multilabel Textual Emotion Classification in Bangla by Combining Transformers with Lexicon Features** (first author).
Bangla Language Processing Workshop (BLP), EMNLP 2023 (PDF)

[4]: **BEmoFusionNet: A Deep Learning Approach for Multimodal Emotion Classification in Bangla Social Media Posts**.
IEEE ICCIT 2023 (PDF)

Technical Skills

Languages: Python (research), C/C++ (competitive programming), Swift (production app)

ML & Systems: PyTorch, CUDA, Linux; familiar with Docker, vLLM

Achievements

Awards: DAC Young Fellow, ACM/IEEE Design Automation Conference (2026).

ICPC: Asia West Continent Finalist (2021 & 2022); 22nd place (2022), 25th place (2021).

Meta Hacker Cup: 38th/34,584 in Qualification Round 2021; 116th/12,692 in Round 1.

Codeforces: Candidate Master (Max Rating 2094, Top 1%); 3300+ problems solved; 330+ rated contests.

Leadership

Vice President, CUET Computer Club (Aug 2022 – Apr 2023): Organized workshops, seminars, and programming contests; coordinated annual planning and logistics.